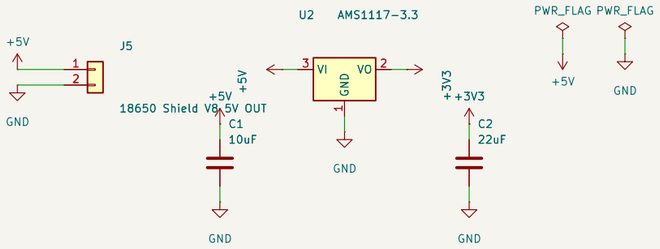
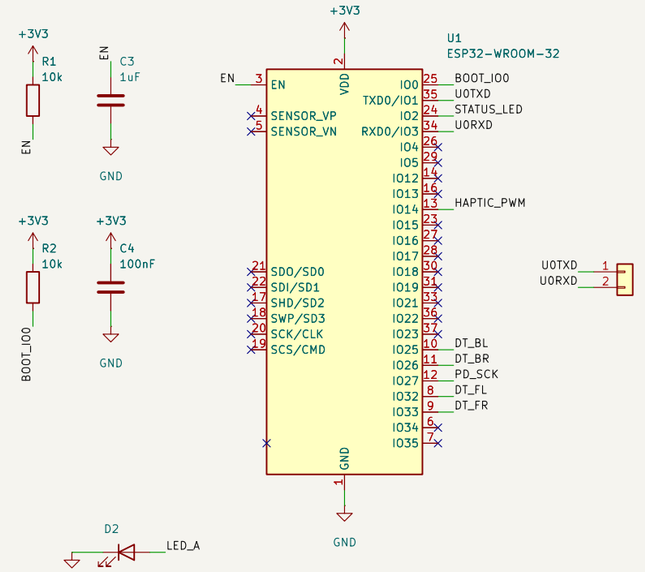
# EquiSense Rev 2.0 — Four-Load-Cell Force Platform (ESP32 + 4x HX711, shared PD_SCK)

Battery-isolated: 2x18650 Shield V8 (5 V) — no mains connection. HX711 boards: RATE (pin 15) tied to DVDD -> 80 SPS; Channel A, gain 128.

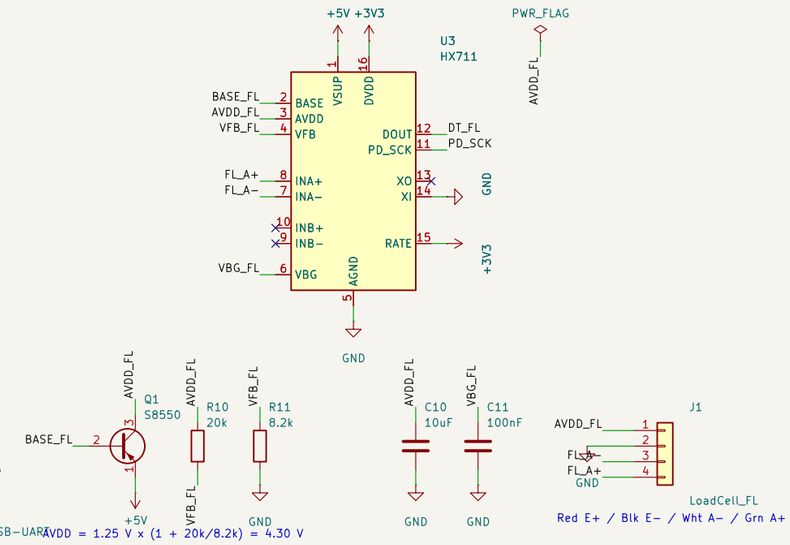
## POWER (isolated)



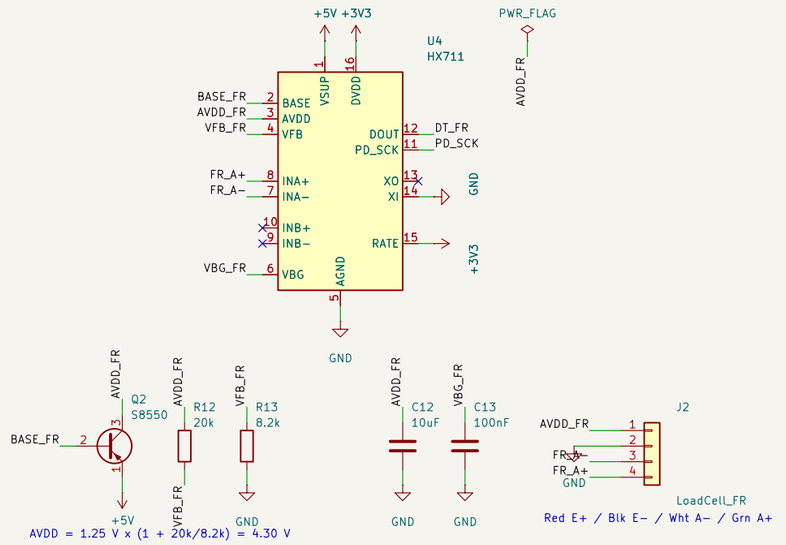

## MCU — ESP32-WROOM-32 (NodeMCU-32S)



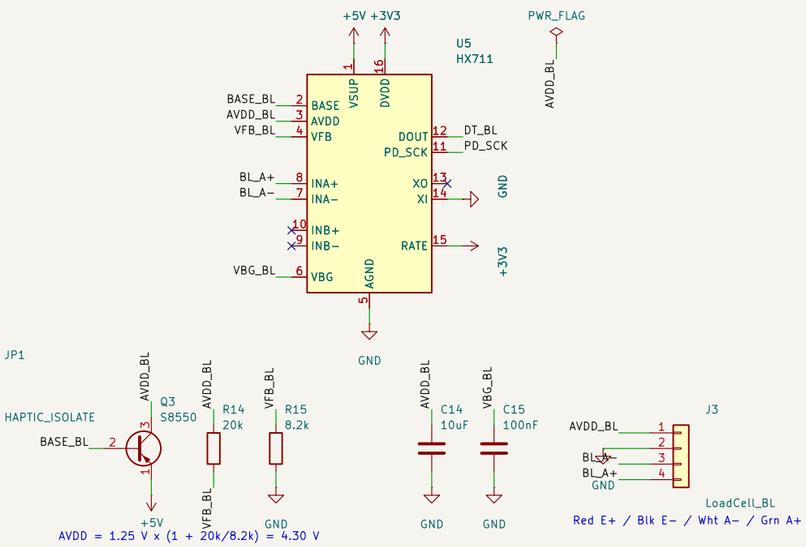

## CH1 FRONT-LEFT (−200,+200) — Phaeton 611E 200 kg, 2.0 mV/V



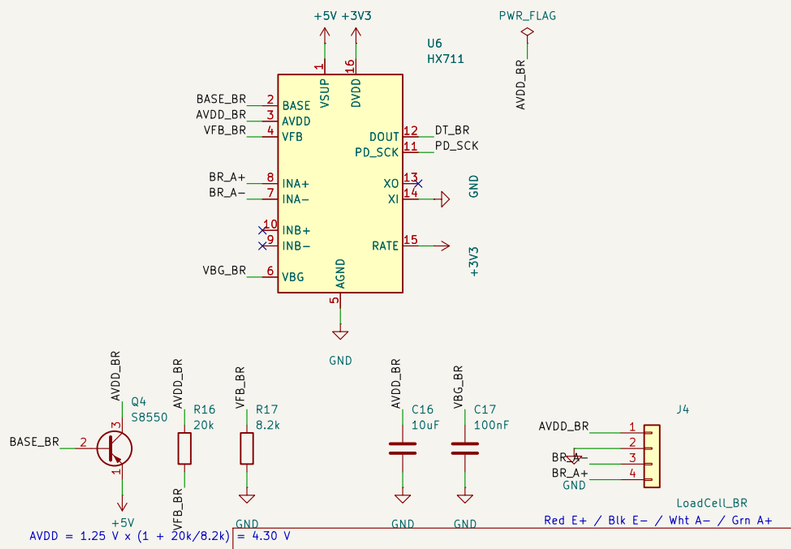

## CH2 FRONT-RIGHT (+200,+200) — Phaeton 611E 200 kg, 2.0 mV/V



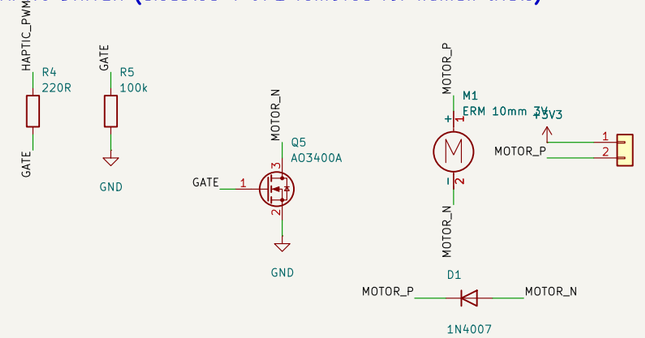

## CH3 BACK-LEFT (−200,−200) — Phaeton 611E 200 kg, 2.0 mV/V




## CH4 BACK-RIGHT (+200,−200) — Phaeton 611E 200 kg, 2.0 mV/V




## HAPTIC DRIVER (disabled + JP1 removed for human trials)




**Dept. of Biomedical Engineering**

Sheet: /
File: EquiSense.kicad_sch

**Title: EquiSense Force Platform — Main Schematic**

| Size: A3 | Date: 2026-10-14 | **Rev: 2.0** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 1/1 |